Supplementary Information for "Efficient methods for *ab initio* modeling of materials using finite-element DFT with τ-mGGA functionals"

Srinibas Nandi[1], Kartick Ramakrishnan[1], Nikhil Kodali[1], Gopalakrishnan Sai Gautam[2], Vishwas Rao[3], Phani Motamarri[1*]

[1*]Department of Computational and Data Sciences, Indian Institute of Science, Bengaluru, 560012, India.

[2]Department of Materials Engineering, Indian Institute of Science, Bengaluru, 560012, India.

[3]Argonne National Laboratory, 9700 S. Cass Avenue, Lemont, IL 60439, USA.

*Corresponding author(s). E-mail(s): phanim@iisc.ac.in;

S1 Projector augmented-wave formulation for τ-mGGA functionals

We extend our previous work[1,2] on the finite-element PAW method for GGA XC functionals to τ-mGGA XC functionals. To this end, we introduce τ^σ as the all-electron kinetic energy which is computed from the pseudo kinetic energy density $\widetilde{\tau}^\sigma$ as

$$\tau^\sigma(\mathbf{x}) = \widetilde{\tau}^\sigma(\mathbf{x}) + \sum_r \sum_a \left[\tau^{a,\sigma}(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r) - \widetilde{\tau}^{a,\sigma}(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r)\right], \tag{S1.1}$$

where the pseudo-kinetic energy $\widetilde{\tau}^\sigma$ is computed from the pseudo wave functions $\{\widetilde{u}^\sigma_{n\boldsymbol{k}}\}$ and fractional occupancy $f^\sigma_{n\boldsymbol{k}}$ as

$$\widetilde{\tau}^\sigma = \frac{1}{2}\sum_n \int_{\mathrm{BZ}} f^\sigma_{n\boldsymbol{k}} \left[\left|\nabla \widetilde{u}^\sigma_{n\boldsymbol{k}}\right|^2 + |\boldsymbol{k}|^2 \left|\widetilde{u}^\sigma_{n\boldsymbol{k}}\right|^2 - i\boldsymbol{k}\cdot\left(\widetilde{u}^{\sigma *}_{n\boldsymbol{k}} \nabla \widetilde{u}^\sigma_{n\boldsymbol{k}} - \widetilde{u}^\sigma_{n\boldsymbol{k}} \nabla \widetilde{u}^{\sigma *}_{n\boldsymbol{k}}\right)\right] d\boldsymbol{k}, \tag{S1.2}$$

while the atom-centered terms $\tau^{a,\sigma}$ and $\widetilde{\tau}^{a,\sigma}$ are given by

$$\tau^{a,\sigma}(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r) := \frac{1}{2}\tau^a_c(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r) + \frac{1}{2}\sum_{\alpha\beta} \nabla\phi^a_\alpha(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r)\cdot\nabla\phi^a_\beta(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r) D^{a,\sigma}_{\alpha\beta} \tag{S1.3}$$

$$\widetilde{\tau}^{a,\sigma}(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r) := \frac{1}{2}\widetilde{\tau}^a_c(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r) + \frac{1}{2}\sum_{\alpha\beta} \nabla\widetilde{\phi}^a_\alpha(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r)\cdot\nabla\widetilde{\phi}^a_\beta(\mathbf{x} - \mathbf{R}^a - \mathbf{L}_r) D^{a,\sigma}_{\alpha\beta}, \tag{S1.4}$$

where $\{D^{a,\sigma}_{\alpha\beta}\}$ are the entries of atom-dependent density matrix whose details are discussed in our previous works[1,2] while, $\{\phi^a_\alpha\}$, $\{\widetilde{\phi}^a_\alpha\}$ are the atom-centered partial wave function and pseudo partial wave functions

respectively in the PAW formalism[3]. The pseudo wave functions $\{\widetilde{u}^{\sigma}_{n\boldsymbol{k}}\}$ are determined from the following governing equations

$$\left(-\frac{1}{2}\nabla^2 - i\boldsymbol{k}\cdot\nabla + \frac{1}{2}|\boldsymbol{k}|^2 + V_{\rm el} + V_{\rm z} + \hat{V}^{\sigma}_{\rm xc} + \mathcal{H}^{\sigma}_{{\rm nloc},\boldsymbol{k}}\right)\widetilde{u}^{\sigma}_{n\boldsymbol{k}} = \epsilon^{\sigma}_{n\boldsymbol{k}}\,(\mathcal{I} + \mathcal{S}_{{\rm nloc},\boldsymbol{k}})\,\widetilde{u}^{\sigma}_{n\boldsymbol{k}},$$

$$1 \le n \le N, \qquad \forall\,\boldsymbol{k} \in {\rm BZ}, \qquad \forall\,\sigma \in \{\uparrow,\downarrow\}$$

$$\hat{V}^{\sigma}_{\rm xc}\,\widetilde{u}^{\sigma}_{n\boldsymbol{k}} = \left[\frac{\partial e_{\rm xc}}{\partial \widetilde{n}^{\sigma}} - \nabla\cdot\left(\frac{\partial e_{\rm xc}}{\partial(\nabla\widetilde{n}^{\sigma})}\right)\right]\widetilde{u}^{\sigma}_{n\boldsymbol{k}}$$

$$-\frac{1}{2}\left[\nabla\cdot\left(\frac{\partial e_{\rm xc}}{\partial\widetilde{\tau}^{\sigma}}\,\nabla\widetilde{u}^{\sigma}_{n\boldsymbol{k}}\right) + \nabla\cdot\left(\frac{\partial e_{\rm xc}}{\partial\widetilde{\tau}^{\sigma}}\,i\boldsymbol{k}\,\widetilde{u}^{\sigma}_{n\boldsymbol{k}}\right) - |\boldsymbol{k}|^2\,\frac{\partial e_{\rm xc}}{\partial\widetilde{\tau}^{\sigma}}\,u^{\sigma}_{n\boldsymbol{k}} + \frac{\partial e_{\rm xc}}{\partial\widetilde{\tau}^{\sigma}}\,i\boldsymbol{k}\cdot\nabla\widetilde{u}^{\sigma}_{n\boldsymbol{k}}\right]$$

$$\mathcal{H}^{\sigma}_{{\rm nloc},\boldsymbol{k}}\,\widetilde{u}^{\sigma}_{n\boldsymbol{k}}(\mathbf{x}) = \sum_{a\in\Omega_p}\sum_{\alpha\beta}\left\{\sum_{r} e^{-i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)}\,\widetilde{p}^{a}_{\alpha}(\mathbf{x}-(\mathbf{R}_a+\mathbf{L}_r))\right\}\Delta h^{a,\sigma}_{\alpha\beta}$$

$$\times\left\{\int_{\Omega_p}\sum_{r'} e^{i\boldsymbol{k}\cdot(\mathbf{x}'-\mathbf{L}_{r'})}\,\widetilde{p}^{a}_{\alpha}(\mathbf{x}'-(\mathbf{R}_a+\mathbf{L}_{r'}))\,\widetilde{u}^{\sigma}_{n\boldsymbol{k}}(\mathbf{x}')\,d\mathbf{x}'\right\}$$

$$\mathcal{S}_{{\rm nloc},\boldsymbol{k}}\,\widetilde{u}^{\sigma}_{n\boldsymbol{k}}(\mathbf{x}) = \sum_{a\in\Omega_p}\sum_{\alpha\beta}\left\{\sum_{r} e^{-i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)}\,\widetilde{p}^{a}_{\alpha}(\mathbf{x}-(\mathbf{R}_a+\mathbf{L}_r))\right\}\Delta s^{a}_{\alpha\beta}$$

$$\times\left\{\int_{\Omega_p}\sum_{r'} e^{i\boldsymbol{k}\cdot(\mathbf{x}'-\mathbf{L}_{r'})}\,\widetilde{p}^{a}_{\alpha}(\mathbf{x}'-(\mathbf{R}_a+\mathbf{L}_{r'}))\,\widetilde{u}^{\sigma}_{n\boldsymbol{k}}(\mathbf{x}')\,d\mathbf{x}'\right\}$$

$$\widetilde{n}^{\sigma} = \frac{1}{2}\widetilde{n}_c + \sum_{n}\fint_{\rm BZ} f^{\sigma}_{n\boldsymbol{k}}\,|\widetilde{u}^{\sigma}_{n\boldsymbol{k}}|^2\,d\boldsymbol{k}, \qquad \widetilde{n} = \widetilde{n}^{\uparrow} + \widetilde{n}^{\downarrow}$$

$$\widetilde{\tau}^{\sigma} = \frac{1}{2}\widetilde{\tau}_c + \frac{1}{2}\sum_{n}\fint_{\rm BZ} f^{\sigma}_{n\boldsymbol{k}}\left[\left|\nabla\widetilde{u}^{\sigma}_{n\boldsymbol{k}}\right|^2 + |\boldsymbol{k}|^2\left|\widetilde{u}^{\sigma}_{n\boldsymbol{k}}\right|^2 - i\boldsymbol{k}\cdot(\widetilde{u}^{\sigma *}_{n\boldsymbol{k}}\,\nabla\widetilde{u}^{\sigma}_{n\boldsymbol{k}} - \widetilde{u}^{\sigma}_{n\boldsymbol{k}}\,\nabla\widetilde{u}^{\sigma *}_{n\boldsymbol{k}})\right]d\boldsymbol{k}$$

$$\widehat{U}^{a,\boldsymbol{k}\sigma}_{\alpha i} := \int_{\Omega_p}\sum_{r'} e^{\mathbf{i}\boldsymbol{k}\cdot(\mathbf{y}-\mathbf{L}_{r'})}\widetilde{p}^{a}_{\alpha}(\mathbf{y}-(\mathbf{R}^a+L_{r'}))\widetilde{u}^{\sigma}_{i,\boldsymbol{k}}(\mathbf{y})d\mathbf{y},\; D^{a\sigma}_{\alpha\beta} = \sum_{i}\fint_{BZ} f^{\sigma}_{n\boldsymbol{k}}\widehat{U}^{a,\boldsymbol{k}\sigma *}_{\alpha i}\widehat{U}^{a,\boldsymbol{k}\sigma}_{\beta n}d\boldsymbol{k}$$

$$\widetilde{b}(\mathbf{x}) = \sum_{a}\sum_{r}\kappa^{a}\widetilde{g}^{a}_{0}(\mathbf{x}-\mathbf{R}^a-\mathbf{L}_r) + \sum_{lm,\alpha\beta}\Delta^{a}_{lm\alpha\beta}D^{a,{\rm tot}}_{\alpha\beta}\widetilde{g}^{a}_{lm}(\mathbf{x}-\mathbf{R}^a-\mathbf{L}_r)$$

$$-\frac{1}{4\pi}\nabla^2 V_{\rm el} = \widetilde{n} + \widetilde{b}, \quad V_z(\mathbf{x}) = \sum_{r}\sum_{a} V^{a}_{z}(\mathbf{x}-\mathbf{R}^a-\mathbf{L}_r)$$

$$f^{\sigma}_{n\boldsymbol{k}} = \left[1+\exp\left(\frac{\epsilon^{\sigma}_{n\boldsymbol{k}}-\mu}{k_BT}\right)\right]^{-1}, \qquad \sum_{\sigma}\sum_{n}\fint_{\rm BZ} f^{\sigma}_{n\boldsymbol{k}}\,d\boldsymbol{k} = N_v$$

The various terms involved in this set of governing equations are explained in detail in our previous works[1], in this section we take a closer look at $\Delta h^{a,\sigma}_{\alpha\beta} = \Delta\mathcal{T}^{a}_{\alpha\beta} + \Delta\mathcal{C}^{a}_{\alpha\beta} + \Delta v^{a,\sigma}_{\rm xc} - \Delta v^{a}_{z} + 2\sum_{ij}\Delta\mathcal{C}^{a}_{ij\alpha\beta}(D^{a\uparrow}_{ij} + D^{a,\downarrow}_{ij})$ where we focus in particular on $\Delta v^{a\sigma}_{\rm xc}$ which has to be extended to support τ-mGGA functionals. To this end, we first expand Eqn. (S1.3) as follows

$$\tau^{a,\sigma}(\mathbf{r}) = \frac{1}{2}\sum_{\alpha\beta}\left[\frac{d\phi^{a}_{\alpha}(r)}{dr}\frac{d\phi^{a}_{\beta}(r)}{dr}\,Y_{l_\alpha m_\alpha}Y_{l_\beta m_\beta}\right.$$

$$+\frac{\phi^a_\alpha(r)\phi^a_\beta(r)}{r^2}\left(\frac{\partial Y_{l_\alpha m_\alpha}}{\partial\theta}\frac{\partial Y_{l_\beta m_\beta}}{\partial\theta}+\frac{1}{\sin^2\theta}\frac{\partial Y_{l_\alpha m_\alpha}}{\partial\phi}\frac{\partial Y_{l_\beta m_\beta}}{\partial\phi}\right)\Bigg]D^{a,\sigma}_{\alpha\beta}+\frac{\tau_c(r)}{2}, \tag{S1.5}$$

where $\boldsymbol{r}$ refers to the spherical coordinate (r,θ,ϕ). Similarly, $\widetilde{\tau}^a(\boldsymbol{r})$ can be evaluated on the radial grid. Finally, $\Delta v^a_{\rm xc}$ is evaluated as

$$\begin{aligned}
\Delta v^{a,\sigma}_{{\rm xc},\alpha\beta} &= \left(\int_{\Omega_a}\frac{\partial\epsilon^a_{\rm xc}[n^{a,\sigma},\nabla n^{a,\sigma},\tau^{a,\sigma}]}{\partial n^{a,\sigma}(r,\widehat{\boldsymbol{\theta}})}\frac{\partial n^{a\sigma}}{\partial D^{a,\sigma}_{\alpha\beta}}r^2drd\widehat{\boldsymbol{\theta}}\right.\\
&\quad\left.-\int_{\Omega_a}\frac{\partial\epsilon^a_{\rm xc}[\widetilde{n}^{a,\sigma},\nabla\widetilde{n}^{a,\sigma},\widetilde{\tau}^{a,\sigma}]}{\partial\widetilde{n}^{a,\sigma}(\widehat{\boldsymbol{\theta}})}\frac{\partial\widetilde{n}^{a\sigma}}{\partial D^{a,\sigma}_{\alpha\beta}}r^2drd\widehat{\boldsymbol{\theta}}\right)\\
&+\left(\int_{\Omega_a}\frac{\partial\epsilon^a_{\rm xc}[n^{a,\sigma},\nabla n^{a,\sigma},\tau^{a,\sigma}]}{\partial\nabla n^{a,\sigma}(r,\widehat{\boldsymbol{\theta}})}\cdot\frac{\partial\nabla n^{a,\sigma}(r,\widehat{\boldsymbol{\theta}})}{\partial D^{a,\sigma}_{\alpha\beta}}r^2drd\widehat{\boldsymbol{\theta}}\right.\\
&\quad\left.-\int_{\Omega_a}\frac{\partial\epsilon^a_{\rm xc}[\widetilde{n}^{a,\sigma},\nabla\widetilde{n}^{a,\sigma},\widetilde{\tau}^{a,\sigma}]}{\partial\nabla\widetilde{n}^{a,\sigma}(\widehat{\boldsymbol{\theta}})}\cdot\frac{\partial\nabla\widetilde{n}^{a,\sigma}(r,\widehat{\boldsymbol{\theta}})}{\partial D^{a,\sigma}_{\alpha\beta}}r^2drd\widehat{\boldsymbol{\theta}}\right)\\
&\frac{1}{2}\int_{\Omega_a}\frac{\delta\epsilon_{\rm xc}}{\delta\tau^{a,\sigma}}\left[\frac{d\phi^a_\alpha(r)}{dr}\frac{d\phi^a_\beta(r)}{dr}Y_{l_\alpha m_\alpha}Y_{l_\beta m_\beta}\right]r^2dr\,d\omega\\
&+\frac{1}{2}\int_{\Omega_a}\frac{\delta\epsilon_{\rm xc}}{\delta\tau^{a,\sigma}}\left[\phi^a_\alpha(r)\,\phi^a_\beta(r)\,\boldsymbol{\Psi}_{l_\alpha m_\alpha}\cdot\boldsymbol{\Psi}_{l_\beta m_\beta}\right]dr\,d\omega
\end{aligned} \tag{S1.6}$$

S2 Stationary properties of the generalized energy functional

Consider the generalized functional

$$\begin{aligned}
\widetilde{E}[\{\widetilde{V}^\sigma\},\{\widetilde{V}^\sigma_\tau\},\{\widetilde{\rho}^\sigma\},\{\widetilde{\tau}^\sigma\},\{\widetilde{f}^\sigma_{n\boldsymbol{k}}\}] &= \sum_\sigma\sum_n\fint_{\rm BZ}\widetilde{f}^\sigma_{n\boldsymbol{k}}\,\epsilon^\sigma_{n\boldsymbol{k}}[\widetilde{V}^\sigma,\widetilde{V}^\sigma_\tau]\,d\boldsymbol{k}-\sum_\sigma\int_{\Omega_p}\left(\widetilde{V}^\sigma\widetilde{\rho}^\sigma+\widetilde{V}^\sigma_\tau\widetilde{\tau}^\sigma\right)d\mathbf{x}\\
&\quad+E_{\rm el}[\widetilde{\rho}]+E_{\rm xc}[\{\widetilde{\rho}^\sigma+\tfrac{1}{2}\rho_{\rm c}\},\{\widetilde{\tau}^\sigma+\tfrac{1}{2}\tau_{\rm c}\}]\\
&\quad+E_{\rm ent}[\{\widetilde{f}^\sigma_{n\boldsymbol{k}}\}]+\mu\left(N_v-\sum_\sigma\sum_n\fint_{\rm BZ}\widetilde{f}^\sigma_{n\boldsymbol{k}}\,d\boldsymbol{k}\right).
\end{aligned} \tag{S2.1}$$

The functional derivatives of $\widetilde{E}$ with respect to each of the variational fields are

$$\frac{\delta\widetilde{E}}{\delta\widetilde{V}^\sigma(\mathbf{x})}=\rho^\sigma[\widetilde{V}^\sigma,\widetilde{V}^\sigma_\tau]-\widetilde{\rho}^\sigma, \tag{S2.2}$$

$$\frac{\delta\widetilde{E}}{\delta\widetilde{V}^\sigma_\tau(\mathbf{x})}=\tau^\sigma[\widetilde{V}^\sigma,\widetilde{V}^\sigma_\tau]-\widetilde{\tau}^\sigma, \tag{S2.3}$$

$$\frac{\delta\widetilde{E}}{\delta\widetilde{\rho}^\sigma(\mathbf{x})}=V^\sigma[\widetilde{\rho}^\sigma+\tfrac{1}{2}\rho_{\rm c},\,\widetilde{\tau}^\sigma+\tfrac{1}{2}\tau_{\rm c}]-\widetilde{V}^\sigma, \tag{S2.4}$$

$$\frac{\delta\widetilde{E}}{\delta\widetilde{\tau}^\sigma(\mathbf{x})}=V^\sigma_\tau[\widetilde{\rho}^\sigma+\tfrac{1}{2}\rho_{\rm c},\,\widetilde{\tau}^\sigma+\tfrac{1}{2}\tau_{\rm c}]-\widetilde{V}^\sigma_\tau, \tag{S2.5}$$

$$\frac{\delta\widetilde{E}}{\delta\widetilde{f}^\sigma_{n\boldsymbol{k}}}=\epsilon^\sigma_{n\boldsymbol{k}}[\widetilde{V}^\sigma,\widetilde{V}^\sigma_\tau]-\mu+k_BT\ln\frac{\widetilde{f}^\sigma_{n\boldsymbol{k}}}{1-\widetilde{f}^\sigma_{n\boldsymbol{k}}}, \tag{S2.6}$$

where we have defined the following quantities. $\rho^\sigma[\widetilde{V}^\sigma,\widetilde{V}^\sigma_\tau]$ and $\tau^\sigma[\widetilde{V}^\sigma,\widetilde{V}^\sigma_\tau]$ denote the spin-resolved electron density and kinetic energy density computed from the wave functions obtained as the solution of the eigenvalue problem defined by Eq. (25) of the main text. We also define

$$V^\sigma[\widetilde{\rho}^\sigma+\tfrac{1}{2}\rho_{\rm c},\,\widetilde{\tau}^\sigma+\tfrac{1}{2}\tau_{\rm c}]=V_{\rm el}[\widetilde{\rho}]+V^\sigma_{\rm xc}[\widetilde{\rho}^\sigma+\tfrac{1}{2}\rho_{\rm c},\,\widetilde{\tau}^\sigma+\tfrac{1}{2}\tau_{\rm c}], \tag{S2.7}$$

$$V_\tau^\sigma[\widetilde{\rho}^\sigma + \tfrac{1}{2}\rho_c,\, \widetilde{\tau}^\sigma + \tfrac{1}{2}\tau_c] = \left.\frac{\partial e_{xc}}{\partial \tau^\sigma}\right|_{\widetilde{\rho}^\sigma + \frac{1}{2}\rho_c,\, \widetilde{\tau}^\sigma + \frac{1}{2}\tau_c}, \tag{S2.8}$$

where $V_{el}[\widetilde{\rho}]$ is the electrostatic potential obtained as the solution of

$$-\frac{1}{4\pi}\nabla^2 V_{el} = \widetilde{\rho} + g, \tag{S2.9}$$

with $\widetilde{\rho} = \sum_\sigma \widetilde{\rho}^\sigma$ left unaugmented (the electrostatic term is bilinear in the density, and the core charge's electrostatic self-interaction is already accounted for within the pseudopotential construction), and V_{xc}^σ is the XC potential given by

$$V_{xc}^\sigma[\widetilde{\rho}^\sigma + \tfrac{1}{2}\rho_c,\, \widetilde{\tau}^\sigma + \tfrac{1}{2}\tau_c] = \left.\frac{\delta E_{xc}}{\delta \widetilde{\rho}^\sigma}\right|_{\widetilde{\rho}^\sigma + \frac{1}{2}\rho_c,\, \widetilde{\tau}^\sigma + \frac{1}{2}\tau_c}. \tag{S2.10}$$

Setting the functional derivatives in Eqs. (S2.2)–(S2.6) to zero and performing simple algebraic manipulations, we obtain from Eqs. (S2.2) and (S2.3)

$$\rho^\sigma[V^\sigma, V_\tau^\sigma] = \widetilde{\rho}^\sigma, \tag{S2.11}$$
$$\tau^\sigma[V^\sigma, V_\tau^\sigma] = \widetilde{\tau}^\sigma, \tag{S2.12}$$

and from Eqs. (S2.4) and (S2.5)

$$\widetilde{V}^\sigma = V^\sigma[\widetilde{\rho}^\sigma, \widetilde{\tau}^\sigma], \tag{S2.13}$$
$$\widetilde{V}_\tau^\sigma = V_\tau^\sigma[\widetilde{\rho}^\sigma, \widetilde{\tau}^\sigma]. \tag{S2.14}$$

Substituting Eqs. (S2.13) and (S2.14) into Eqs. (S2.11) and (S2.12) yields the self-consistency conditions

$$\rho^\sigma[V^\sigma[\widetilde{\rho}^\sigma, \widetilde{\tau}^\sigma],\, V_\tau^\sigma[\widetilde{\rho}^\sigma, \widetilde{\tau}^\sigma]] = \widetilde{\rho}^\sigma, \tag{S2.15}$$
$$\tau^\sigma[V^\sigma[\widetilde{\rho}^\sigma, \widetilde{\tau}^\sigma],\, V_\tau^\sigma[\widetilde{\rho}^\sigma, \widetilde{\tau}^\sigma]] = \widetilde{\tau}^\sigma, \tag{S2.16}$$

or, equivalently,

$$F[(\{\rho^\sigma\}, \{\tau^\sigma\})] = (\{\rho^\sigma\}, \{\tau^\sigma\}), \tag{S2.17}$$

which is precisely the fixed-point condition defining the solution of the Kohn-Sham eigenvalue problem in the τ-mGGA formalism. Furthermore, Eq. (S2.6) set to zero gives the Fermi-Dirac occupation

$$\widetilde{f}_{n\boldsymbol{k}}^\sigma = \frac{1}{1 + \exp\left(\frac{\epsilon_{n\boldsymbol{k}}^\sigma - \mu}{k_B T}\right)}, \tag{S2.18}$$

consistent with the finite-temperature Kohn-Sham formulation. Thus, the generalized functional defined in Eq. (24) of the main text has a stationary point at the solution of the Kohn-Sham τ-mGGA eigenvalue problem.

Since ρ_c, τ_c are prescribed functions of $\mathbf{x}$ at a fixed atomic configuration $\{\mathbf{R}_a\}$, they do not vary under the above functional derivatives and only shift the argument at which the XC-derived quantities are evaluated.

S3 Gateaux derivatives of the generalized τ-mGGA functional

We evaluate the directional derivative $d\widetilde{E}_S^\varepsilon/d\varepsilon|_{\varepsilon=0}$ by first writing the parametrized functional in the perturbed space as

$$\widetilde{E}_S^\varepsilon[\{V^{\sigma\varepsilon}\}, \{V_\tau^{\sigma\varepsilon}\}, \{\rho^{\sigma\varepsilon}\}, \{\tau^{\sigma\varepsilon}\}, \{f_{n\boldsymbol{k}^\varepsilon}^{\sigma\varepsilon}\}]$$

$$
\begin{aligned}
&= \sum_\sigma \sum_n \fint_{\mathrm{BZ}^\varepsilon} f^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}\, \epsilon^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}[V^{\sigma\varepsilon}, V^{\sigma\varepsilon}_\tau]\, d\boldsymbol{k}^\varepsilon - \sum_\sigma \int (V^{\sigma\varepsilon}\rho^{\sigma\varepsilon} + V^{\sigma\varepsilon}_\tau \tau^{\sigma\varepsilon})\, d\mathbf{x}^\varepsilon \\
&\quad + E^\varepsilon_{\mathrm{el}}[\rho^\varepsilon] + E^\varepsilon_{\mathrm{xc}}\big[\{\rho^{\sigma\varepsilon} + \tfrac{1}{2}\rho^\varepsilon_{\mathrm{c}}\}, \{\tau^{\sigma\varepsilon} + \tfrac{1}{2}\tau^\varepsilon_{\mathrm{c}}\}\big] \\
&\quad + E^\varepsilon_{\mathrm{ent}}[\{f^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}\}] + \mu\left(N_v - \sum_\sigma \sum_n \fint_{\mathrm{BZ}^\varepsilon} f^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}\, d\boldsymbol{k}^\varepsilon\right).
\end{aligned} \tag{S3.1}
$$

Since all variational derivatives vanish at the stationary point, the total derivative reduces to a partial derivative at fixed ground-state fields

$$
\left.\frac{d\widetilde{E}^\varepsilon_S}{d\varepsilon}\right|_{\varepsilon=0} = \left.\frac{\partial\, \widetilde{E}^\varepsilon_S[V^\sigma, \{V^\sigma_\tau\}, \{\rho^\sigma\}, \{\tau^\sigma\}, \{f^\sigma_{n\boldsymbol{k}}\}]}{\partial\varepsilon}\right|_{\varepsilon=0}. \tag{S3.2}
$$

Transforming the integrals back to the unperturbed domain yields

$$
\begin{aligned}
&\widetilde{E}^\varepsilon_S[V^\sigma, \{V^\sigma_\tau\}, \{\rho^\sigma\}, \{\tau^\sigma\}, \{f^\sigma_{n\boldsymbol{k}}\}] \\
&= \sum_\sigma \sum_n \fint_{\mathrm{BZ}} f^\sigma_{n\boldsymbol{k}}\, \epsilon^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}[V^\sigma, V^\sigma_\tau]\, d\boldsymbol{k} - \sum_\sigma \int_{\Omega_p} (V^\sigma\rho^\sigma + V^\sigma_\tau\tau^\sigma) \det\left(\frac{\partial \mathbf{x}^\varepsilon}{\partial \mathbf{x}}\right) d\mathbf{x} \\
&\quad + E^\varepsilon_{\mathrm{el}}[\rho, \mathbf{R}^\varepsilon] + E^\varepsilon_{\mathrm{xc}}[\{\rho^\sigma_{\mathrm{xc}}\}, \{\tau^\sigma_{\mathrm{xc}}\}] + E_{\mathrm{ent}}[\{f^\sigma_{n\boldsymbol{k}}\}] + \mu\left(N_v - \sum_\sigma \sum_n \fint_{\mathrm{BZ}} f^\sigma_{n\boldsymbol{k}}\, d\boldsymbol{k}\right).
\end{aligned} \tag{S3.3}
$$

Using the standard relations

$$
\left.\frac{\partial}{\partial\varepsilon} \det\left(\frac{\partial \mathbf{x}^\varepsilon}{\partial \mathbf{x}}\right)\right|_{\varepsilon=0} = \boldsymbol{I} : \nabla\boldsymbol{\Upsilon}, \qquad \left.\frac{\partial}{\partial\varepsilon}\left(\frac{\partial \mathbf{x}}{\partial \mathbf{x}^\varepsilon}\right)\right|_{\varepsilon=0} = -\nabla\boldsymbol{\Upsilon}, \tag{S3.4}
$$

the partial derivative becomes

$$
\begin{aligned}
\left.\frac{\partial\, \widetilde{E}^\varepsilon_S}{\partial\varepsilon}\right|_{\varepsilon=0} &= \sum_\sigma \sum_n \fint_{\mathrm{BZ}} f^\sigma_{n\boldsymbol{k}} \left.\frac{\partial\, \epsilon^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}[V^\sigma, V^\sigma_\tau]}{\partial\varepsilon}\right|_{\varepsilon=0} d\boldsymbol{k} \\
&\quad - \sum_\sigma \int_{\Omega_p} (V^\sigma\rho^\sigma + V^\sigma_\tau\tau^\sigma)\, \boldsymbol{I} : \nabla\boldsymbol{\Upsilon}\, d\mathbf{x} + \left.\frac{\partial\, E^\varepsilon_{\mathrm{el}}}{\partial\varepsilon}\right|_{\varepsilon=0} + \left.\frac{\partial\, E^\varepsilon_{\mathrm{xc}}}{\partial\varepsilon}\right|_{\varepsilon=0}.
\end{aligned} \tag{S3.5}
$$

The XC term in Eq. (S3.5) carries two distinct contributions, namely the passive response of E_{xc} to the deformation of the domain and the explicit $\{\mathbf{R}_a\}$-dependence introduced by the core fields $\rho^\varepsilon_{\mathrm{c}}, \tau^\varepsilon_{\mathrm{c}}$. To separate them, we expand $\partial E^\varepsilon_{\mathrm{xc}}/\partial\varepsilon$ in terms of its arguments $\rho^\sigma_{\mathrm{xc}}, \tau^\sigma_{\mathrm{xc}}$

$$
\left.\frac{\partial E^\varepsilon_{\mathrm{xc}}}{\partial\varepsilon}\right|_{\varepsilon=0} = \left.\frac{\partial E^\varepsilon_{\mathrm{xc}}[\rho^\sigma_{\mathrm{xc}}, \tau^\sigma_{\mathrm{xc}}]}{\partial\varepsilon}\right|_{\substack{\rho^\sigma_{\mathrm{xc}}, \tau^\sigma_{\mathrm{xc}} \\ \text{fixed}}} + \sum_\sigma \int_{\Omega_p} \frac{\delta E_{\mathrm{xc}}}{\delta\rho^\sigma_{\mathrm{xc}}} \left.\frac{d\rho^{\sigma\varepsilon}_{\mathrm{xc}}}{d\varepsilon}\right|_{\varepsilon=0} d\mathbf{x} + \sum_\sigma \int_{\Omega_p} \frac{\delta E_{\mathrm{xc}}}{\delta\tau^\sigma_{\mathrm{xc}}} \left.\frac{d\tau^{\sigma\varepsilon}_{\mathrm{xc}}}{d\varepsilon}\right|_{\varepsilon=0} d\mathbf{x}. \tag{S3.6}
$$

The first term, holding $\rho^\sigma_{\mathrm{xc}}, \tau^\sigma_{\mathrm{xc}}$ fixed and differentiating only through the Jacobian and the deformed gradient, gives $\mathbf{E}_3$ [Eq. (S3.17)].

The remaining two terms are evaluated at the frozen ground-state fields of Eq. (S3.3), where ρ^σ and τ^σ carry no ε-dependence. The core fields do, however, since $\rho_{\mathrm{c}}, \tau_{\mathrm{c}}$ are prescribed by the atomic positions and are therefore transported by Γ^ε, so that

$$
\left.\frac{d\rho^{\sigma\varepsilon}_{\mathrm{xc}}}{d\varepsilon}\right|_{\varepsilon=0} = \frac{1}{2}\left.\frac{d\rho^\varepsilon_{\mathrm{c}}}{d\varepsilon}\right|_{\varepsilon=0}, \qquad \left.\frac{d\tau^{\sigma\varepsilon}_{\mathrm{xc}}}{d\varepsilon}\right|_{\varepsilon=0} = \frac{1}{2}\left.\frac{d\tau^\varepsilon_{\mathrm{c}}}{d\varepsilon}\right|_{\varepsilon=0}. \tag{S3.7}
$$

With V^σ_{xc} the multiplicative XC potential of Eq. (S2.10), evaluated at the ground state, the last two terms of Eq. (S3.6) then reduce to

$$
\frac{1}{2}\sum_\sigma \int_{\Omega_p} V^\sigma_{\mathrm{xc}} \left.\frac{d\rho^\varepsilon_{\mathrm{c}}}{d\varepsilon}\right|_{\varepsilon=0} d\mathbf{x} + \frac{1}{2}\sum_\sigma \int_{\Omega_p} \frac{\partial e_{\mathrm{xc}}}{\partial\tau^\sigma_{\mathrm{xc}}} \left.\frac{d\tau^\varepsilon_{\mathrm{c}}}{d\varepsilon}\right|_{\varepsilon=0} d\mathbf{x} = F^{\mathrm{NLCC}}, \tag{S3.8}
$$

with the core-field derivatives evaluated in Section S4. This is the sole origin of the NLCC force appearing in Eq. (27).

Considering the first term in Eq. (S3.5), we need to evaluate the derivative of the eigenvalues in the perturbed space

$$\frac{\partial\,\epsilon^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}[\,V^\sigma,\,V^\sigma_\tau\,]}{\partial\varepsilon} = \frac{\partial}{\partial\varepsilon}\left\langle u^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}\,\middle|\,\mathcal{H}^{\sigma\varepsilon}_{\boldsymbol{k}^\varepsilon}[\,V^\sigma,\,V^\sigma_\tau\,]\,\middle|\,u^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}\right\rangle_\varepsilon . \tag{S3.9}$$

Invoking the Hellmann-Feynman theorem, this reduces to

$$\left.\frac{\partial\,\epsilon^{\sigma\varepsilon}_{n\boldsymbol{k}^\varepsilon}}{\partial\varepsilon}\right|_{\varepsilon=0} = \left\langle u^\sigma_{n\boldsymbol{k}}\,\middle|\,\left.\frac{\partial\,\mathcal{H}^{\sigma\varepsilon}_{\boldsymbol{k}^\varepsilon}[\,V^\sigma,\,V^\sigma_\tau\,]}{\partial\varepsilon}\right|_{\varepsilon=0}\,\middle|\,u^\sigma_{n\boldsymbol{k}}\right\rangle . \tag{S3.10}$$

The Hamiltonian derivative decomposes into local and nonlocal contributions

$$\frac{\partial\,\mathcal{H}^{\sigma\varepsilon}_{\boldsymbol{k}^\varepsilon}}{\partial\varepsilon} = \frac{\partial\,\mathcal{H}^{\sigma,\varepsilon}_{\mathrm{loc},\boldsymbol{k}^\varepsilon}}{\partial\varepsilon} + \frac{\partial\,\mathcal{H}^{\varepsilon}_{\mathrm{nloc},\boldsymbol{k}^\varepsilon}}{\partial\varepsilon}. \tag{S3.11}$$

For the local part, the expectation value in the perturbed space can be written as

$$\begin{aligned}\left\langle u^\sigma_{n\boldsymbol{k}}\,\middle|\,\left.\frac{\partial\,\mathcal{H}^{\sigma,\varepsilon}_{\mathrm{loc},\boldsymbol{k}^\varepsilon}[V^\sigma,V^\sigma_\tau]}{\partial\varepsilon}\right|_{\varepsilon=0}\,\middle|\,u^\sigma_{n\boldsymbol{k}}\right\rangle = \frac{\partial}{\partial\varepsilon}\Bigg\{\int_{\Omega_p}\Bigg(&-\frac{u^{\sigma*}_{n\boldsymbol{k}}\,\nabla^2_\varepsilon\,u^\sigma_{n\boldsymbol{k}}}{2} - u^{\sigma*}_{n\boldsymbol{k}}\,i\boldsymbol{k}^\varepsilon\cdot\nabla_\varepsilon\,u^\sigma_{n\boldsymbol{k}} \\ &+ \frac{|\boldsymbol{k}^\varepsilon|^2}{2}\,|u^\sigma_{n\boldsymbol{k}}|^2 + (V^\sigma + V^\varepsilon_{\mathrm{loc}} - V^\varepsilon_g)\,|u^\sigma_{n\boldsymbol{k}}|^2 \\ &- \frac{1}{2}\Big[u^{\sigma*}_{n\boldsymbol{k}}\,\nabla_\varepsilon\cdot\Big(V^\sigma_\tau\,\nabla_\varepsilon\,u^\sigma_{n\boldsymbol{k}}\Big) + u^{\sigma*}_{n\boldsymbol{k}}\,\nabla_\varepsilon\cdot\Big(V^\sigma_\tau\,i\boldsymbol{k}^\varepsilon\,u^\sigma_{n\boldsymbol{k}}\Big) \\ &- |\boldsymbol{k}^\varepsilon|^2\,V^\sigma_\tau\,|u^\sigma_{n\boldsymbol{k}}|^2 + V^\sigma_\tau\,i\,u^{\sigma*}_{n\boldsymbol{k}}\,\boldsymbol{k}^\varepsilon\cdot\nabla_\varepsilon\,u^\sigma_{n\boldsymbol{k}}\Big]\Bigg)\,d\mathbf{x}\Bigg\}\Bigg|_{\varepsilon=0}.\end{aligned} \tag{S3.12}$$

The nonlocal Hamiltonian derivative does not involve τ^σ, and its contribution is given explicitly in Eq. (S3.18).

Invoking the Laplacian relation for the τ-mGGA Hamiltonian,

$$\begin{aligned}\frac{1}{2}\nabla^2 u^\sigma_{n\boldsymbol{k}} = &\left(\frac{|\boldsymbol{k}|^2}{2} - i\boldsymbol{k}\cdot\nabla + (V^\sigma + V_{\mathrm{loc}} - V_g) + \mathcal{H}^\sigma_{\mathrm{nloc},\boldsymbol{k}} - \epsilon^\sigma_{n\boldsymbol{k}}\right) u^\sigma_{n\boldsymbol{k}} \\ &- \frac{1}{2}\Big(\nabla\cdot(V^\sigma_\tau\nabla u^\sigma_{n\boldsymbol{k}}) + \nabla\cdot(V^\sigma_\tau\,i\boldsymbol{k}\,u^\sigma_{n\boldsymbol{k}}) - |\boldsymbol{k}|^2\,V^\sigma_\tau\,u^\sigma_{n\boldsymbol{k}} + V^\sigma_\tau\,i\boldsymbol{k}\cdot\nabla u^\sigma_{n\boldsymbol{k}}\Big).\end{aligned} \tag{S3.13}$$

The electrostatic energy has the local variational form

$$E_{\mathrm{el}}[\rho,\{\mathbf{R}_a\}] = \max_{\widehat{V}_{\mathrm{el}}}\left\{\int_{\Omega_p}\left[-\frac{|\nabla\widehat{V}_{\mathrm{el}}|^2}{8\pi} + (\rho+g)\widehat{V}_{\mathrm{el}}\right]d\mathbf{x}\right\} - C_{\mathrm{el}}(\{\mathbf{R}_a\}). \tag{S3.14}$$

The maximizing field is V_{el}. Its implicit variation does not contribute because Eq. (S3.14) is stationary with respect to $\widehat{V}_{\mathrm{el}}$, and hence

$$\left.\frac{dE^\varepsilon_{\mathrm{el}}}{d\varepsilon}\right|_{\varepsilon=0} = \int_{\Omega_p}\mathbf{E}_2 : \nabla\boldsymbol{\Upsilon}\,d\mathbf{x} + F^{\mathrm{sm}} + F^C. \tag{S3.15}$$

Together with the eigenvalue and XC terms, Eq. (S3.15) gives Eq. (27).

Remaining force contributions

The terms $\mathbf{E}_2$, $F^{\mathrm{psp,nloc}}$, $F^{\mathrm{ext,corr}}$, F^{sm}, and F^C do not involve the kinetic energy density τ^σ, and $\mathbf{E}_3$ depends on it only through e_{xc} and $\partial e_{\mathrm{xc}}/\partial(\nabla\rho^\sigma_{\mathrm{xc}})$. Following the electrostatic treatment in Eqs. (7)–(9),

their derivation proceeds as in the GGA case[4] for the spin-polarized collinear setting. These terms are given below.

The electrostatic Eshelby tensor $\mathbf{E}_2$ is given by

$$\mathbf{E}_2 = \left(\rho\, V_{\mathrm{el}} - \frac{|\nabla V_{\mathrm{el}}|^2}{8\pi}\right) \boldsymbol{I} + \frac{1}{4\pi}\, \nabla V_{\mathrm{el}} \otimes \nabla V_{\mathrm{el}}, \tag{S3.16}$$

where $\rho = \sum_\sigma \rho^\sigma$. The XC Eshelby tensor $\mathbf{E}_3$ is

$$\mathbf{E}_3 = e_{\mathrm{xc}}\, \boldsymbol{I} - \sum_\sigma \frac{\partial e_{\mathrm{xc}}}{\partial(\nabla \rho^\sigma_{\mathrm{xc}})} \otimes \nabla \rho^\sigma_{\mathrm{xc}}. \tag{S3.17}$$

The nonlocal pseudopotential force is given by $F^{\mathrm{psp,nloc}} = F^{\dagger}_{\mathrm{nloc}} + F_{\mathrm{nloc}}$, where

$$\begin{aligned} F_{\mathrm{nloc}} = \sum_\sigma \sum_{a\in\Omega_p} \sum_{\beta,\beta'} \sum_r \sum_n \fint_{\mathrm{BZ}} f^\sigma_{n\boldsymbol{k}} \left[\int_{\Omega_p} u^{\sigma *}_{n\boldsymbol{k}}(\mathbf{x})\, e^{-i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)}\, p^a_\beta(\mathbf{x}-\mathbf{L}_r-\mathbf{R}_a)\, d\mathbf{x}\right] \\ \times D^a_{\beta\beta'} \int_{\Omega_p} \sum_{r'} e^{i\boldsymbol{k}\cdot(\mathbf{x}'-\mathbf{L}_{r'})}\, p^a_{\beta'}(\mathbf{x}'-\mathbf{L}_{r'}-\mathbf{R}_a) \\ \times \Big[-(\boldsymbol{\Upsilon}(\mathbf{x}') - \boldsymbol{\Upsilon}(\mathbf{R}_a+\mathbf{L}_{r'}))\cdot \nabla u^\sigma_{n\boldsymbol{k}}(\mathbf{x}') - i\boldsymbol{k}\cdot\boldsymbol{\Upsilon}(\mathbf{R}_a)\, u^\sigma_{n\boldsymbol{k}}(\mathbf{x}')\Big]\, d\mathbf{x}'\, d\boldsymbol{k}. \end{aligned} \tag{S3.18}$$

The external correction term is

$$F^{\mathrm{ext,corr}} = \sum_{a\in\Omega_p} \sum_r \int_{\Omega_p} \rho(\mathbf{x}) \nabla\big[V^a_{\mathrm{loc}}(|\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r|) - V^a_g(\mathbf{x},\mathbf{R}_a+\mathbf{L}_r)\big]\cdot(\boldsymbol{\Upsilon}(\mathbf{x}) - \boldsymbol{\Upsilon}(\mathbf{R}_a+\mathbf{L}_r))\, d\mathbf{x}. \tag{S3.19}$$

Since V^a_{loc} and V^a_g depend on $\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r$, Eq. (S3.19) follows from the chain rule. The smooth nuclear charge contribution is

$$F^{\mathrm{sm}} = -\sum_{a\in\Omega_p} \sum_r \int_{\Omega_p} g^a(\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r)\, \nabla V_{\mathrm{el}} \cdot (\boldsymbol{\Upsilon}(\mathbf{x}) - \boldsymbol{\Upsilon}(\mathbf{R}_a+\mathbf{L}_r))\, d\mathbf{x}. \tag{S3.20}$$

The analytic correction contributes

$$F^C = -\frac{\partial C^\varepsilon_{\mathrm{el}}(\{\mathbf{R}^\varepsilon_a\})}{\partial \varepsilon}\bigg|_{\varepsilon=0}. \tag{S3.21}$$

The terms $F^{\mathrm{ext,corr}}$, F^{sm}, and F^C involve only the total density $\rho = \sum_\sigma \rho^\sigma$ and the spin-independent electrostatic and pseudopotential fields. The nonlocal pseudopotential coefficients $D^a_{\beta\beta'}$ in Eq. (S3.18) are scalar with respect to spin in the spin-polarized collinear case, in contrast to the 2×2 spin matrices appearing in the noncollinear formulation of ref. 4.

S4 Derivatives of the core fields

The core fields $\rho_{\mathrm{c}}, \tau_{\mathrm{c}}$ of Eq. (5) are rigid templates centered on the nuclei, so the configurational map Γ^ε carries both the field point and the template center, $\mathbf{R}^\varepsilon_a = \Gamma^\varepsilon(\mathbf{R}_a)$. The perturbed core density is therefore

$$\rho^\varepsilon_{\mathrm{c}}(\mathbf{x}) = \sum_{a\in\Omega_p} \sum_r \rho^a_{\mathrm{c}}\left(\left|\Gamma^\varepsilon(\mathbf{x}) - \Gamma^\varepsilon(\mathbf{R}_a+\mathbf{L}_r)\right|\right), \tag{S4.1}$$

and identically for $\tau_{\mathrm{c}}^{\varepsilon}$. Differentiating at $\varepsilon = 0$ with $d\Gamma^{\varepsilon}(\mathbf{x})/d\varepsilon|_0 = \boldsymbol{\Upsilon}(\mathbf{x})$ gives

$$\left.\frac{d\rho_{\mathrm{c}}^{\varepsilon}}{d\varepsilon}\right|_{\varepsilon=0} = \sum_{a\in\Omega_p}\sum_{r} \nabla\rho_{\mathrm{c}}^{a}(|\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r|)\cdot\big(\boldsymbol{\Upsilon}(\mathbf{x})-\boldsymbol{\Upsilon}(\mathbf{R}_a+\mathbf{L}_r)\big), \tag{S4.2}$$

$$\left.\frac{d\tau_{\mathrm{c}}^{\varepsilon}}{d\varepsilon}\right|_{\varepsilon=0} = \sum_{a\in\Omega_p}\sum_{r} \nabla\tau_{\mathrm{c}}^{a}(|\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r|)\cdot\big(\boldsymbol{\Upsilon}(\mathbf{x})-\boldsymbol{\Upsilon}(\mathbf{R}_a+\mathbf{L}_r)\big). \tag{S4.3}$$

Since only the difference $\boldsymbol{\Upsilon}(\mathbf{x}) - \boldsymbol{\Upsilon}(\mathbf{R}_a + \mathbf{L}_r)$ appears, a uniform translation $\boldsymbol{\Upsilon} \equiv \boldsymbol{c}$ makes Eqs. (S4.2)–(S4.3) vanish pointwise, as required. The accompanying functional derivatives follow from the chain rule on $\rho_{\mathrm{xc}}^{\sigma} = \rho^{\sigma} + \frac{1}{2}\rho_{\mathrm{c}}$ and $\tau_{\mathrm{xc}}^{\sigma} = \tau^{\sigma} + \frac{1}{2}\tau_{\mathrm{c}}$,

$$\frac{\delta E_{\mathrm{xc}}}{\delta\rho_{\mathrm{c}}} = \frac{1}{2}\sum_{\sigma} V_{\mathrm{xc}}^{\sigma}(\mathbf{x}), \qquad \frac{\delta E_{\mathrm{xc}}}{\delta\tau_{\mathrm{c}}} = \frac{1}{2}\sum_{\sigma} \frac{\partial e_{\mathrm{xc}}}{\partial\tau_{\mathrm{xc}}^{\sigma}}(\mathbf{x}), \tag{S4.4}$$

with V_{xc}^{σ} the multiplicative XC potential of Eq. (S2.10), evaluated at the ground state. The second derivative is purely multiplicative because $\tau_{\mathrm{xc}}^{\sigma}$ enters e_{xc} algebraically. Substituting Eqs. (S4.2)–(S4.4) into Eq. (S3.8) yields the NLCC force $F^{\mathrm{NLCC}} = F_{\rho}^{\mathrm{NLCC}} + F_{\tau}^{\mathrm{NLCC}}$ of Eq. (27), with

$$F_{\rho}^{\mathrm{NLCC}} = \frac{1}{2}\sum_{\sigma}\sum_{a\in\Omega_p}\sum_{r}\int_{\Omega_p} V_{\mathrm{xc}}^{\sigma}(\mathbf{x})\,\nabla\rho_{\mathrm{c}}^{a}(|\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r|)\cdot\big(\boldsymbol{\Upsilon}(\mathbf{x})-\boldsymbol{\Upsilon}(\mathbf{R}_a+\mathbf{L}_r)\big)\,d\mathbf{x}, \tag{S4.5}$$

$$F_{\tau}^{\mathrm{NLCC}} = \frac{1}{2}\sum_{\sigma}\sum_{a\in\Omega_p}\sum_{r}\int_{\Omega_p} \frac{\partial e_{\mathrm{xc}}}{\partial\tau_{\mathrm{xc}}^{\sigma}}(\mathbf{x})\,\nabla\tau_{\mathrm{c}}^{a}(|\mathbf{x}-\mathbf{R}_a-\mathbf{L}_r|)\cdot\big(\boldsymbol{\Upsilon}(\mathbf{x})-\boldsymbol{\Upsilon}(\mathbf{R}_a+\mathbf{L}_r)\big)\,d\mathbf{x}. \tag{S4.6}$$

S5 Atomic forces and unit cell stress

This section gives the atomic forces and the unit cell stress that follow from Eq. (27), in the form used in the implementation. The notation follows the main text. Define $\boldsymbol{\xi}_{ar}(\mathbf{x}) = \mathbf{x} - \mathbf{R}_a - \mathbf{L}_r$ and the projector contractions

$$\mathcal{P}_{a\beta}^{\sigma n\boldsymbol{k}} = \int_{\Omega_p}\sum_{r} e^{i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)} p_{\beta}^{a}(\boldsymbol{\xi}_{ar}) u_{n\boldsymbol{k}}^{\sigma}\,d\mathbf{x}, \quad \mathcal{G}_{a\beta}^{\sigma n\boldsymbol{k}} = \int_{\Omega_p}\sum_{r} e^{i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)} p_{\beta}^{a}(\boldsymbol{\xi}_{ar}) \nabla u_{n\boldsymbol{k}}^{\sigma}\,d\mathbf{x}. \tag{S5.1}$$

Atomic forces are obtained by choosing a generator localized about atom A. With the unit cell and spatial discretization fixed, differentiation with respect to $\mathbf{R}_A$, followed by integration by parts, gives

$$\frac{\partial\mathcal{P}_{A\beta}^{\sigma n\boldsymbol{k}}}{\partial\mathbf{R}_A} = \mathcal{G}_{A\beta}^{\sigma n\boldsymbol{k}} + i\boldsymbol{k}\,\mathcal{P}_{A\beta}^{\sigma n\boldsymbol{k}}. \tag{S5.2}$$

Thus, differentiating $E_{\mathrm{nloc}} = \sum_{\sigma n a}\sum_{\beta,\beta'} \fint_{\mathrm{BZ}} f_{n\boldsymbol{k}}^{\sigma} \mathcal{P}_{a\beta}^{\sigma n\boldsymbol{k}*} D_{\beta\beta'}^{a} \mathcal{P}_{a\beta'}^{\sigma n\boldsymbol{k}} d\boldsymbol{k}$ gives the nonlocal contribution to the atomic force. The atomic force $\mathbf{f}_A = -\partial E/\partial\mathbf{R}_A$ is

$$\begin{aligned}\mathbf{f}_A = -2\,\mathrm{Re}\Bigg[\sum_{\sigma}\sum_{n}\sum_{\beta}\fint_{\mathrm{BZ}} f_{n\boldsymbol{k}}^{\sigma}\left(\sum_{\beta'} D_{\beta\beta'}^{A}\mathcal{P}_{A\beta'}^{\sigma n\boldsymbol{k}}\right)^{*}\left(\mathcal{G}_{A\beta}^{\sigma n\boldsymbol{k}} + i\boldsymbol{k}\,\mathcal{P}_{A\beta}^{\sigma n\boldsymbol{k}}\right)d\boldsymbol{k}\Bigg] \\ -\sum_{r}\int_{\Omega_p}\Big(\nabla\rho(\mathbf{x})\left[V_{\mathrm{loc}}^{A}(|\boldsymbol{\xi}_{Ar}|) - V_{g}^{A}(\mathbf{x},\mathbf{R}_A+\mathbf{L}_r)\right] + g^{A}(\boldsymbol{\xi}_{Ar})\nabla V_{\mathrm{el}}(\mathbf{x})\Big)d\mathbf{x} + \frac{\partial C_{\mathrm{el}}}{\partial\mathbf{R}_A} + \mathbf{f}_A^{\mathrm{NLCC}},\end{aligned} \tag{S5.3}$$

where the NLCC contribution, including both the core density and core kinetic energy density, is

$$\mathbf{f}_A^{\mathrm{NLCC}} = \frac{1}{2}\sum_{\sigma}\sum_{r}\int_{\Omega_p}\left[V_{\mathrm{xc}}^{\sigma}(\mathbf{x})\,\nabla\rho_{\mathrm{c}}^{A}(|\boldsymbol{\xi}_{Ar}|) + \frac{\partial e_{\mathrm{xc}}}{\partial\tau_{\mathrm{xc}}^{\sigma}}(\mathbf{x})\,\nabla\tau_{\mathrm{c}}^{A}(|\boldsymbol{\xi}_{Ar}|)\right]d\mathbf{x}, \tag{S5.4}$$

where V_{xc}^{σ} is the multiplicative XC potential of Eq. (S2.10), evaluated at the ground state. The nonlocal and local-pseudopotential terms in Eq. (S5.3) are written after integration by parts. For the latter,

$\int_{\Omega_p} \rho\, \nabla(V_{\rm loc}^A - V_g^A) d\mathbf{x} = -\int_{\Omega_p} \nabla\rho\, (V_{\rm loc}^A - V_g^A) d\mathbf{x}$. The signs follow from $\mathbf{f}_A = -\partial E/\partial \mathbf{R}_A$ and the occurrence of $-C_{\rm el}$ in the energy.

For the unit cell stress, choose the affine generator $\boldsymbol{\Upsilon}(\mathbf{x}) = \boldsymbol{C}\mathbf{x}$, where $\boldsymbol{C}$ is a constant 3×3 matrix. Then $\nabla\boldsymbol{\Upsilon} = \boldsymbol{C}$ and $\boldsymbol{k}^\varepsilon = (\boldsymbol{I} - \varepsilon \boldsymbol{C}^T)\boldsymbol{k}$. Let $\eta^\sigma = \partial e_{\rm xc}/\partial \tau_{\rm xc}^\sigma$ and define the local $\boldsymbol{k}$-variation contribution

$$\mathbf{S}^{K,\rm loc} = \mathrm{Re}\Bigg[\sum_\sigma \sum_n \fint_{\rm BZ} f_{n\boldsymbol{k}}^\sigma \int_{\Omega_p} \Big\{ (1+\eta^\sigma)\left[i u_{n\boldsymbol{k}}^{\sigma *}\, \boldsymbol{k} \otimes \nabla u_{n\boldsymbol{k}}^\sigma - |u_{n\boldsymbol{k}}^\sigma|^2\, \boldsymbol{k}\otimes\boldsymbol{k}\right] - \frac{i}{2}\eta^\sigma\, \boldsymbol{k} \otimes (u_{n\boldsymbol{k}}^\sigma \nabla u_{n\boldsymbol{k}}^{\sigma *} + u_{n\boldsymbol{k}}^{\sigma *} \nabla u_{n\boldsymbol{k}}^\sigma) \Big\} d\mathbf{x}\, d\boldsymbol{k} \Bigg]. \tag{S5.5}$$

This term is the affine specialization of the local part of F^K in Eq. (30) and accounts for the variation of $\boldsymbol{k}$ under a cell deformation.

The nonlocal stress contribution is written using

$$\mathbf{X}_{a\beta}^{\sigma n\boldsymbol{k}} = \int_{\Omega_p} \sum_r e^{i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)} p_\beta^a(\boldsymbol{\xi}_{ar}) u_{n\boldsymbol{k}}^\sigma\, \boldsymbol{\xi}_{ar}\, d\mathbf{x}, \quad \mathbf{M}_{a\beta}^{\sigma n\boldsymbol{k}} = \int_{\Omega_p} \sum_r e^{i\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{L}_r)} p_\beta^a(\boldsymbol{\xi}_{ar}) \nabla u_{n\boldsymbol{k}}^\sigma \otimes \boldsymbol{\xi}_{ar}\, d\mathbf{x}, \tag{S5.6}$$

as

$$\mathbf{S}^{\rm psp,nloc} = 2\,\mathrm{Re}\Bigg\{ \sum_\sigma \sum_{a\in\Omega_p} \sum_n \sum_\beta \fint_{\rm BZ} f_{n\boldsymbol{k}}^\sigma \left(\sum_{\beta'} D_{\beta\beta'}^a \mathcal{P}_{a\beta'}^{\sigma n\boldsymbol{k}} \right)^* \left(\mathbf{M}_{a\beta}^{\sigma n\boldsymbol{k}} - i\mathbf{X}_{a\beta}^{\sigma n\boldsymbol{k}} \otimes \boldsymbol{k} \right) d\boldsymbol{k} \Bigg\}^T. \tag{S5.7}$$

For this affine generator, the local-pseudopotential, smooth nuclear charge, and NLCC contributions are

$$\mathbf{S}^{\rm ext,corr} = \sum_{a\in\Omega_p} \sum_r \int_{\Omega_p} \rho(\mathbf{x}) \nabla\left[V_{\rm loc}^a(|\boldsymbol{\xi}_{ar}|) - V_g^a(\mathbf{x}, \mathbf{R}_a + \mathbf{L}_r) \right] \otimes \boldsymbol{\xi}_{ar}\, d\mathbf{x}, \tag{S5.8}$$

$$\mathbf{S}^{\rm sm} = -\sum_{a\in\Omega_p} \sum_r \int_{\Omega_p} g^a(\boldsymbol{\xi}_{ar}) \nabla V_{\rm el}(\mathbf{x}) \otimes \boldsymbol{\xi}_{ar}\, d\mathbf{x}, \tag{S5.9}$$

$$\mathbf{S}^{\rm NLCC} = \frac{1}{2} \sum_\sigma \sum_{a\in\Omega_p} \sum_r \int_{\Omega_p} \left[V_{\rm xc}^\sigma(\mathbf{x})\, \nabla \rho_{\rm c}^a(|\boldsymbol{\xi}_{ar}|) + \eta^\sigma(\mathbf{x})\, \nabla \tau_{\rm c}^a(|\boldsymbol{\xi}_{ar}|) \right] \otimes \boldsymbol{\xi}_{ar}\, d\mathbf{x}. \tag{S5.10}$$

The sum of $\mathbf{S}^{\rm ext,corr}$ and the $(V_{\rm loc} - V_g)\rho\boldsymbol{I}$ term in $\int_{\Omega_p} \mathbf{E}_1 d\mathbf{x}$ can equivalently be evaluated as

$$-\sum_{a\in\Omega_p} \sum_r \int_{\Omega_p} \nabla\rho(\mathbf{x}) \left[V_{\rm loc}^a(|\boldsymbol{\xi}_{ar}|) - V_g^a(\mathbf{x}, \mathbf{R}_a + \mathbf{L}_r) \right] \otimes \boldsymbol{\xi}_{ar}\, d\mathbf{x}. \tag{S5.11}$$

This identity follows by integration by parts. The boundary terms cancel for periodic systems and vanish for non-periodic systems. With the affine generator $\boldsymbol{\Upsilon} = \boldsymbol{C}\mathbf{x}$, i.e. the deformation $\mathbf{x} \mapsto (\boldsymbol{I} + \varepsilon \boldsymbol{C})\mathbf{x}$, Eq. (27) gives the unit cell stress of Eq. (31) as

$$\boldsymbol{\sigma} = \frac{1}{|\Omega_p|} \left[\int_{\Omega_p} (\mathbf{E} + \mathbf{E}_\tau)\, d\mathbf{x} + \mathbf{S}^{K,\rm loc} + \mathbf{S}^{\rm psp,nloc} + \mathbf{S}^{\rm ext,corr} + \mathbf{S}^{\rm sm} + \mathbf{S}^{\rm NLCC} - \frac{\partial C_{\rm el}^{\boldsymbol{C}}(\{\mathbf{R}_a^{\boldsymbol{C}}\})}{\partial \boldsymbol{C}} \Big|_{\boldsymbol{C}=\mathbf{0}} \right]. \tag{S5.12}$$

Here $C_{\rm el}^{\boldsymbol{C}}$ is Eq. (9) evaluated in the affinely deformed cell with $\mathbf{R}_a^{\boldsymbol{C}} = (\boldsymbol{I} + \boldsymbol{C})\mathbf{R}_a$ and $\mathbf{L}_r^{\boldsymbol{C}} = (\boldsymbol{I} + \boldsymbol{C})\mathbf{L}_r$. Equations (S5.3) and (S5.12) reduce to the GGA expressions when $\partial e_{\rm xc}/\partial \tau_{\rm xc}^\sigma = 0$ and $\tau_{\rm c} = 0$.

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